LESSON 2.1
FINDING THE GREATEST COMMON FACTOR

LESSON INTRODUCTION

MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.

LEARNING TARGETS

- I can find the greatest common factor (GCF) in mathematical and real-world contexts.
- I can rewrite fractions in simplest form using the greatest common factor (GCF) of the numerator and denominator.

BENCHMARK COHERENCE

BUILDING ON

MA.4.AR.3.1 Determine factor pairs for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither.

ESSENTIAL QUESTIONS

- What is a greatest common factor?
- What strategies can be used to find the greatest common factor of two whole numbers?
- How can the greatest common factor be used to rewrite a fraction in simplest form?

LESSON NARRATIVE

Students extend prior knowledge of finding factors of whole numbers to introduce how to find the greatest common factor (GCF) of two whole numbers. The lesson begins with an anchor task to introduce the concepts and definitions within a real-world context. Students then use the GCF to transition from informal "simplifying" to formally rewriting equivalent expressions, focusing on why the original fraction and the "simplest" form fraction are equivalent. Each new concept is followed by a brief Try It that students can do independently or with a partner.

COMPONENT SUMMARY

2.1.1 WARM-UP *(~5 MIN.)*

Math Symbols Video

2.1.2 GUIDED INSTRUCTION *(10-15 MIN.)*

Using a table to identify GCF and factor pairs

2.1.3 TRY IT *(5-10 MIN.)*

Finding GCF within a real-world context

2.1.4 GUIDED INSTRUCTION *(10-15 MIN.)*

GCF in the context of numerator and denominator for reducing fractions

2.1.5 TRY IT *(5-10 MIN.)*

Simplifying and explaining equivalent expressions

2.1.6 YOUR TURN! *(5-10 MIN.)*

Simplifying fractions using GCF or common factors

2.1.7 WRAP-UP *(~5 MIN.)*

Application problems, one skill-based and one within real-world context

RESOURCES

GLOSSARY

Key Terms:

- Factor
- Greatest common factor (GCF)

Additional Terms:

- N/A

MATERIALS

- (Optional) Manipulatives for 2.1.2 Guided Instruction

ADDITIONAL PREPARATION

- 2.1.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may get stuck on finding factors for larger numbers.
- Students may not recognize that they may not need to find all factors for both numbers to find GCF.
- Students may not recognize that when dividing a numerator and denominator by the GCF, the only remaining common factor is 1.

THINGS TO CONSIDER

- Consider making a word wall or anchor chart for vocabulary words introduced in this lesson, including GCF and simplest form, and to review any related previously learned vocabulary words. Add to this vocabulary chart after multiples are introduced later in Lesson 2, which can help clarify the difference between factors and multiples and minimize confusion.
- Consider reviewing other strategies, including using organized lists or using divisibility rules to identify factor pairs.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

2.1.2 and 2.1.4 Guided Instructions give students an opportunity to consider different real-world contexts in which the GCF is used. Consider giving students the opportunity to think of another situation in which the GCF is used to make equal sizes of groups throughout the lesson, then pair with a partner to share.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students will find the GCF of two numbers within real-world contexts in the both the Guided Instruction and Try It portions of this lesson. Consider highlighting the connection of GCF to everyday experiences using a Think-Pair-Share. Other possible real-world contexts could include packaging, making teams, and grouping items.

DIFFERENTIATE INSTRUCTION

Consider finding the GCF of smaller numbers during Try It as an entry point for students who are still building fluency with recognizing factor pairs and common factors.

Consider using manipulatives to represent “snacks” and “letters” while completing the table from 2.1.2 Guided Instruction to help students visualize and physically create equal-sized groups using the GCF.

If time permits, in the Wrap-Up consider having students create their own version of a real-world context problem that includes using the GCF to create equal-sized groups.

2.1.1 WARM-UP: WHERE DO MATH SYMBOLS COME FROM?

Component Overview: *Display the "Where Do Math Symbols Come From?" video for the entire class. After students watch this two-minute video, prompt the students to read and answer the questions in the Warm-Up independently or in partners.*

TEACHER MOVES

After the Warm-Up, the teacher can have a few students share symbols they wrote for question 1 or have students share their answers to question 3 to build on the understanding of why symbols are useful in math. The discussions should be brief and not take time away from core instruction.

QUESTIONING

To increase engagement, consider having students record the math symbols while watching the video, categorizing them into symbols they know and symbols they do not know.

2.1.1 Warm-Up: Where Do Math Symbols Come From?

Math is full of symbols that include lines, dots, arrows, English letters, Greek letters, and more. It can look and sometimes feel like its own language, and in many ways it is!

Robert Recorde invented the equals sign, =, because he was tired of writing "is equal to" over and over again.

1. **What is a word or phrase that you use a symbol to represent instead of writing it out each time?**
   Answers will vary. I use "&" to represent "and." Instead of writing "percent," I use the percent symbol (%).

2. **Why did early mathematicians decide to use symbols to express operations, phrases, and mathematical ideas?**
   Answers will vary. Mathematicians may have wanted to avoid using a lot of words, especially when sharing ideas with mathematicians who speak different languages. Symbols were used to organize mathematical ideas in a clear way.

3. **Why are math symbols still helpful to us today?**
   Answers will vary. Math symbols help us express mathematical ideas across language barriers. Symbols reduce the amount of writing that is needed.

2.1.2 Guided Instruction: Greatest Common Factor

1. Marcus and Tonya are making care packages for soldiers overseas. Inside of each package, they will include popular snacks and handwritten letters. They have 16 snacks and 12 letters to put in the care packages. They want each care package to be equally sized, containing both snacks and letters, with nothing left over.

   a. Complete the tables below to show the possible combinations when making each care package.

Distributing Snacks Equally	
Number of Care Packages	Number of Snacks per Package
1	16
2	8
4	4
8	2
16	1

Distributing Letters Equally	
Number of Care Packages	Number of Letters per Package
1	12
2	6
3	4
4	3
6	2
12	1

   b. What is the greatest number of care packages that can be created using **only** snacks?
   16 care packages

   c. What is the greatest number of care packages that can be created using **only** letters?
   12 care packages

   d. What is the greatest number of care packages that can be created using **both** snacks and letters?
   4 care packages

2.1.2 GUIDED INSTRUCTION: GREATEST COMMON FACTOR

Component Overview: *Students will be exploring an anchor task to find the GCF in the real-world context of creating care packages. Using a table to organize factor pairs, students will see common factors and be guided to identify the greatest number of care packages that is created using both snacks and letters. Guided Instruction positions the teacher to "think aloud" as students begin to formalize a conceptual understanding of GCFs. The teacher should identify and formalize the definitions of factor and GCF during this activity.*

DIFFERENTIATED INSTRUCTION

Consider using manipulatives to represent the snacks and letters. Place 16 "snacks" and 12 "letters" on a display or table visible to all learners, then physically move the manipulatives to make equal-sized groups as learners complete the table. Emphasize that equal groups are required for the care packages to be complete.

ENRICHMENT

Questions 1a and 1b inform students' understanding of what it means by using 'both' in question 1d.

- How are the numbers in each column for the snacks table related to 16?
- How are the numbers in each column for the letters table related to 12?
- For question 1d: Can 2, 3, 4, 5, 6, 8, 12, or 16 care packages be created using both snacks and letters?

2.1.2 GUIDED INSTRUCTION: GREATEST COMMON FACTOR (CONTINUED)

TEACHER MOVES

Think-Pair-Share

Consider discussing that making care packages is only one real-world context and probe students to think of another scenario in which the GCF is used to make equal-sized groups. Possible other contexts include; making equal-sized groups, packaging, or teams. Emphasize conceptual connections between the real-world contexts requiring the creation of equal-sized groups.

A ***factor*** is a number that divides into another number exactly and without leaving a remainder.

For example, the ***factors*** of 20 are 1, 2, 4, 5, 10, and 20.

The numbers (or expressions) that are being multiplied are called the factors. When numbers (or expressions) are multiplied, the answer is called a product.

factor × factor = product

e. How were the factors of 12 and 16 used in the previous problem?

- The factors of 16 were used to determine the whole number of [o letters / o snacks] that could be included, depending on the number of care packages.
  - snacks
- The factors of 12 were used to determine the whole number of [o letters / o snacks] that could be included, depending on the number of care packages.
  - letters

The ***greatest common factor***, or ***GCF***, is the largest common factor of two or more numbers.

ENRICHMENT

How can you check that all the factor pairs for two numbers have been found?

DIFFERENTIATED INSTRUCTION

For question 2, consider using an organized list (vertical or horizontal) to visually show all factor pairs. To engage visual learners in seeing factor pairs, list the factors from least to greatest and consider using a line to connect the factor pairs. Then circle the common factors between the two lists to help visually identify which is the greatest.

An example of how to find the ***greatest common factor (GCF)*** is shown.

The factors of 15 are 1, 3, 5, and 15.

The factors of 20 are 1, 2, 4, 5, 10, and 20.

The GCF of 15 and 20 is 5.

f. Complete the statement to explain what the GCF of 16 and 12 is from the previous problem.

The GCF of 16 and 12 is __4__, which represents the greatest number of care packages that can be created using
- o only snacks.
- o only letters.
- o both snacks and letters.

2. Find the greatest common factor of 6 and 18.

	Factors	GCF:
6	1,2,3,6	6
18	1,2,3,6,9,18	

2.1.3 Try It: Greatest Common Factor

1. **Find the greatest common factor of each pair of numbers.**

   a. 15 and 200
   GCF = 5

   b. 123 and 41
   GCF = 41

   c. 54 and 36
   GCF = 18

   d. 282 and 108
   GCF = 6

2. **Penelope is organizing the teacher-student relay race. The teams must be divided evenly, with no participants left out. If there are 56 students registered and 21 teachers, what is the greatest number of teacher-student teams that will be able to participate in the relay race?**
   7 teacher-student teams

2.1.3 TRY IT: GREATEST COMMON FACTOR

Component Overview: *Students practice finding the GCF of pairs of numbers by using factor pairs. Question 2 introduces the real-world concept of creating equal-sized teams, extending students' understanding of finding and using GCF in everyday experiences.*

DIFFERENTIATED INSTRUCTION

Consider possible arrangements in the classroom to allow students to work in partners for added peer support or facilitate student agency by allowing students to choose to complete two or three of the four total questions. For movement, post each of these to a corner of the classroom and have students walk around to each corner while finding the GCF.

TEACHER MOVES

For questions 1a-d, consider asking students to discuss whether they need to find all factors to find the GCF. For example, in question 1a when finding the GCF of 15 and 200, they do not need to list every factor of 200. They could list the factors of 15 (a much shorter list) and from there determine the greatest of those that is also a factor of 200.

ENRICHMENT

For question 2, what do you notice about the real-world context that helps you know we are looking for the GCF of these values?

2.1.4 GUIDED INSTRUCTION: WRITING EQUIVALENT FRACTIONS USING GCF

Component Overview: *Students review the key vocabulary of equivalent fractions and are introduced to using the GCF to rewrite fractions in simplest form. Students focus on reasoning related to identifying equivalent fractions and explaining that they are equivalent because they have the same value. Consider using this sample as a guide for a "think aloud" as the teacher, or place students in pairs and have them use the self-guided prompts to explore the process of GCF through the lens of fractions.*

ENRICHMENT

What would be the new GCF of the remaining values? What does that tell you about when you know an answer is in "simplest" form?

QUESTIONING

For students who might finish early or have extra time while other students are still formalizing the concept, consider having students create their own example and switch with a partner to simplify using the GCF.

2.1.4 Guided Instruction: Writing Equivalent Fractions Using GCF

The GCF can also be used to rewrite fractions. When using the GCF to write an equivalent fraction, the result is often said to be in "simplest" form.

To **simplify a fraction**, generate an equivalent fraction by dividing the numerator and denominator by the **GCF** of the numerator and denominator.

1. In the fraction $\frac{25}{90}$, the numerator is 25 and the denominator is 90.

   a. What is the greatest common factor of 25 and 90?
   GCF = 5

   b. Divide the numerator and denominator by the GCF. What is the resulting fraction?
   $\frac{5}{18}$

   c. Complete the statement to describe the relationship between the fractions.

   After dividing the numerator and denominator of $\frac{25}{90}$ by their GCF, the resulting fraction, $\boxed{\frac{5}{18}}$, ○ is / ○ is not equivalent to $\frac{25}{90}$, rewritten in simplest form.

2.1.5 Try It: Writing Equivalent Fractions using GCF

1. Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

a. $\frac{8}{28}$

GCF = 4

$\frac{2}{7}$

b. $\frac{45}{90}$

GCF = 45

$\frac{1}{2}$

c. $\frac{16}{34}$

GCF = 2

$\frac{8}{17}$

2. Mikayla states that $\frac{900}{1000}$ is not the same value as $\frac{9}{10}$, because they are fractions of different sizes. Is she correct? Provide an explanation to support your answer.

Answers will vary. She is incorrect. They are equivalent fractions; $\frac{900}{1000}$ can be reduced to $\frac{9}{10}$ by dividing the numerator and denominator by 100.

2.1.5 TRY IT: WRITING EQUIVALENT FRACTIONS USING GCF

Component Overview: *Question 1 gives students the opportunity to (individually or in partners) write equivalent fractions using the GCF of the numerator and denominator. Question 2 focuses on reasoning and explaining why two fractions are equivalent using the GCF as the simplifying factor.*

QUESTIONING

For extension on question 1a and 1b, give students the opportunity to notice the difference between generating equivalent fractions with a common factor that is not the GCF and using the GCF. Emphasize that both approaches are valid but using a common factor may take multiple steps. For example, in 1b, simplifying with the common factors 9 and 5 requires two steps, whereas simplifying with the GCF 45 takes one step.

DIFFERENTIATED INSTRUCTION

For question 2, consider reviewing divisibility rules for numbers ending in zero or multiple zeros. This strategy may help students who are having difficulty identifying common factors for large numbers. Prompting could include, “What do you notice about the ending of 900 and 1000? How can you tell that both numbers are divisible by 10? By 100? What is the greatest common factor: 10 or 100?”

2.1.6 YOUR TURN!

Component Overview: *Students practice both learning targets in these problems. Question 1 focuses on finding the GCF of a pair of numbers, while question 2 uses the GCF to simplify a fraction. Question 3 provides another example of a real-world scenario for using the GCF to create equal-sized groups.*

TEACHER MOVES

Consider engaging students in discussion on how to use multiple approaches to finding GCF, including organized lists, finding factor pairs using known multiplication facts, finding factor pairs for the smaller of the two values first to compare common factors, and/or revisiting divisibility rules to quickly determine if factors are shared. Questioning could include, “How can we find the GCF using a different strategy?” or “Which strategy do you prefer? Why?”

ENRICHMENT

For any part of question 1: What are the factors of __ and __? Which is the greatest common factor? Do I need to find all factors of both numbers to find the GCF?

QUESTIONING

Consider asking students to extend their thinking by drawing a visual model or diagram of the equal-sized groups of the pencils and stickers used for each bag.

2.1.6 Your Turn!

1. **Find the greatest common factor (GCF) of each pair of numbers.**

   a. 18 and 24 — *GCF* = 6

   b. 45 and 60 — *GCF* = 15

   c. 665 and 30 — *GCF* = 5

2. **Rewrite each fraction in simplest form using the GCF of the numerator and denominator.**

   a. $\frac{29}{87}$ — *GCF* = 29, $\frac{1}{3}$

   b. $\frac{10}{15}$ — *GCF* = 5, $\frac{2}{3}$

   c. $\frac{175}{210}$ — *GCF* =35, $\frac{5}{6}$

3. **Mrs. Chaunuan is making gift bags filled with pencils and stickers for class prizes. Mrs. Chaunuan has 24 pencils and 36 stickers to use. Each bag will have the same number of each item, with no items left over. What is the greatest number of gift bags she can make?**

   12 gift bags

2.1.7 Wrap-Up: Ticket Out the Door

1. Rewrite $\frac{52}{104}$ in simplest form.
   GCF $= 52$
   $\frac{1}{2}$

2. Diego is preparing brownies and cookies for a bake sale. He has 48 brownies and 64 cookies. Diego wants to make bake sale bags. Each bag will have the same amount of each item, with no items left over. What is the greatest number of bake sale bags he can make?
   16 *bake sale bags*

2.1.7 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' ability to rewrite fractions in simplest form using the GCF of the numerator and denominator and apply finding the GCF to a real-world context.*

DIFFERENTIATED INSTRUCTION

For visual learners, consider allowing students to represent their thinking for question 2 using a model or diagram. Students could draw or label a model of the brownies and cookies, then explain how they used the GCF to determine their answer.

